

CNN-Based Detector for Spectrum Sensing With General Noise Models

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Abstract—In this paper, we consider spectrum sensing (SS) problems with various general noise models such as Middleton class A (MCA), isometric complex symmetric α -stable ($S\alpha S$), and isometric complex generalized Gaussian distribution (CGGD). This approach enables us to examine the effect of practical phenomena such as impulsive noise on SS problems. In this general framework, we propose a detector based on convolutional neural networks (CNNs) with favorable performance under various noise models. The proposed model-free and data-driven CNN offers robustness in diverse noise scenarios. Thus, it can be utilized in environments with different physical behaviors. We demonstrate this method outperforms the highly regarded likelihood ratio test (LRT) in most cases. For all impulsive cases, the proposed CNN is the superior detector, providing a near-optimum performance for the conventional Gaussian noise. We indicate the proposed data-driven CNN offers an appropriate alternative solution to LRT. However, it requires more computational operations, a rich training dataset, and a training process, instead. Furthermore, the main rationale for proposing this CNN is that it enables the network to generalize its effective performance to various noise models and cases. To this end, quantitative simulations confirm superiority of the proposed CNN compared to other recent deep-learning methods.

Index Terms—Cognitive radio, spectrum sensing, deep learning, CNN, likelihood ratio test, impulsive noise.

I. INTRODUCTION

THE growing demand of users for data communication through a wireless channel is a well-known trend that requires more spectral resources. To serve this demand, CRs enable users to utilize the spectrum dynamically and more efficiently. For efficient exploitation of underutilized spectrum by a CR, SS plays an essential role. In a CR network, a secondary user (SU) is permitted to sense the radio spectrum to find the unused spectrum (spectrum holes) in order to utilize it opportunistically. However, after finding an idle frequency band, SU is mandated to sense the spectrum periodically to vacate that frequency band as quickly as a licensed user, called primary user (PU), becomes active.

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It is obvious that a method of SS with superior performance in the detection of PUs and spectrum holes leads to more efficient spectrum exploitation. Thus, improving SS methods and related detectors has been the interest of numerous researches [1], [2], [3]. On the other hand, it is expected that a wireless SU manages to perform SS successfully in a wide variety of environments with different physical behaviors that affect wireless communications.

Each of these environments inevitably involves various phenomena that cause remarkably diverse effects on SS and also noise model. In the literature, various noise models and their impact on receivers have been studied [4], [5], [6]. In contrast to common noise models in wireless communication, these proposed statistical models are able to include both conventional Gaussian case and also impulsive models with heavy-tailed distributions. Thus, these noise models with the ability to capture impulsive noise, offer generalization for describing the behavior of noise in different environments.

Impulsive noise can be resulted due to atmospheric conditions like lightning or man-made sources such as power lines, cars' engine, and devices like microwave ovens, printers and photocopiers, electrical machines, etc. [7]. Presence of impulsive noise has been reported in numerous works; for instance, Middleton class A and B distributions with the ability to model background noise and several interfering sources have been offered in [4]. Impulses in noise have been measured for both outdoor environments in UHF frequency bands [6] and also in indoor environments [8]. In [9] and [10], $S\alpha S$ model and also MCA have shown better accordance with empirical measurements than Gaussian distribution for various environments. This accordance has been observed with estimated parameters that lead the models to heavy-tailed distributions, e.g., $\alpha < 2$ for $S\alpha S$. Furthermore, $S\alpha S$ and MCA distributions have been validated by empirical measurements for specific environments [9], [10], [11]. MCA and $S\alpha S$ also are physically precise models for the Poisson field of interferers due to co-channel interferers [11], [12]. Generalized Gaussian distribution (GGD) is another non-Gaussian distribution employed particularly for modeling multiple access interference in a channel [13], [14].

SS problem has been investigated under these non-Gaussian models in a variety of works since conventional energy-based detectors cannot handle heavy-tailed noise [9], [15], [16]. Combined with order statistics (COS) detector in [15] and covariate coefficient absolute value (CCAV) detector in [9] have been proposed for non-Gaussian noise model following $S\alpha S$ distribution. Also, a modified energy detector

(ED) and fractional lower-order moments (FLOM) instead of second-order moments (SOMs) such as energy have been offered in [17], [18], and [19], respectively. MCA model for SS has been considered with ED in [20]. Polarity-coincidence-array (PCA) spectrum sensing has been utilized in [21] and [22] for contaminated received signals by noise following GGD.

Many proposed detectors in the literature are based on a statistical model which is assumed to explain the received signal and effect of channel and noise for a specific environment. For instance, LRT has played a key role in the derivation of tests and detectors for hypothesis testing problems such as SS. This widely employed test requires a fixed model and distribution and related likelihoods under distinct hypotheses to provide a solution for hypothesis testing problems. However, recent developments in deep neural networks (DNN) have established novel data-driven and model-free methods for classification problems. Since a DNN-based detector is independent of models, it can offer a general solution for SS under all mentioned noise models with different sets of parameters corresponding to that specific environment.

In recent works, different DNNs have been proposed for SS. In [23], a simple artificial neural network (ANN) with inputted energy of received signals for SS has been proposed. Also, CNN as fusion center with local energy-based decisions of SUs has been used for cooperative SS in [24]. Additionally, CNNs have been employed with inputted spectral correlation density of received signals [25], [26] and with inputted sample covariance matrix (SCM) of received signals on an array of antennas for SS problem, called covariance matrix-based CNN (CM-CNN) [27], [28], [29]. Other types of DNNs like stacked autoencoder (SAE) have been also proposed to use the main samples of the received signal of orthogonal frequency division multiplexing (OFDM) for SS [30].

These papers have mainly focused on SS in the conventional Gaussian model and used second-order statistics which are sensitive to impulsive noise. However, since for $S\alpha S$ distribution, second-order moments do not exist except for special cases, authors of [31] proposed a CNN with inputted covariance matrix which is computed based on FLOMs. In [16], a CNN under non-circular $S\alpha S$ noise model has been proposed for SS with a single antenna which shows robust performance against impulsive noise. In this work, we propose a CNN for a multiantenna scenario with three isometric noise models. Consequently, we indicate the generality of the solution for SS by the proposed network. This generality and robustness are not exclusive just for impulsive or Gaussian noise (in one model). They are also reachable for a higher level where the models of noise are distinctive. We show this by satisfying performance of the proposed DNN-based detector for these three noise models and with their different parameters. The main contributions of this work are as follows

- In this work, we present the signal model and SS problem for multi-antenna SU with different general noise models. In contrast to the previous works which just consider one general noise model, three important isometric complex noise distributions of GGD, MCA, and $S\alpha S$ are considered in this work for SS. Additionally, we derive

LRT for all these three different cases. LRT is beyond a detector and as a test it has been widely used for derivation of detectors for different models. To the best of our knowledge, for the first time, this derivation of LRT under various models enables us to comprehensively compare the highly regarded test of LRT (not just one detector) with DNN-based and data-driven detectors.

- We propose a model-free and data-driven CNN-based detector with the potential of generalization to various models as an alternative to LRT. For complete comparison of this model-based and dominant test with the proposed CNN, LRT is numerically computed even for the cases without a closed-form formula. Simulation results show the proposed CNN outperforms detectors derived based on LRT in different instances of various models due to its generalization ability. We also discuss the training stage, bias adjustment, and the potential of CNN in order to be the maximum a posteriori (MAP) detector on the training dataset. Thus, we conclude that providing a rich training dataset is highly effective for improving the performance of CNN.
- In addition to detectors derived by LRT, we compare the proposed method with several model-based and recent data-driven detectors such as recurrent neural networks (RNNs), ANN, SAE with fully connected (FC) layers, and CNNs in the literature. We show that the proposed CNN outperforms these detectors, and in contrast to its DNN-based counterparts, it is robust against all these three noise models, and it is able to generalize its effective performance to test signals with unseen range of power. These advantages are achieved due to the selected architecture and its appropriate inputted signal.

The generally favorable performance of the proposed detector and its robustness under different models and distributions inside a family with different parameters, in contrast to the well-known test of LRT, is one of the major contributions of this study. Since LRT has been a dominant method for derivation of detectors in various SS problems, this work suggests that data-driven methods with a suitable architecture can be considered as another proper option for SS problems.

The rest of the paper comes as follows: In Section II, signal model and general SS problems are explained. The SS problems for different noise models and their related LRTs are discussed in Section III. Section IV is about the proposed CNN and its training stage. In Section V, simulation results are presented, and finally, Section VI includes the conclusion of this work.

II. SIGNAL MODEL

In this work, we consider an SU with multiple receiving antennas to sense the spectrum for finding spectrum holes. For modeling SS problem, we take into account two hypotheses. For the alternate hypothesis, denoted by H_1 , a single-antenna PU transmits through a channel sensed by an SU. Consequently, the SU receives the following equivalent baseband samples of the transmitted signal

$$r_m[n] = h_m x[n] + w_m[n], \quad (1)$$

with $1 \leq m \leq M$ and $1 \leq n \leq N$, where M and N denote the number of receiving antennas and number of sensed samples, respectively. Here, $x[n]$ is the n -th sample of the transmitted signal by the PU. h_m is the fading coefficient of the wireless channel between PU's single antenna and the m -th antenna of the SU. The wireless channel is modeled by flat fading, and it is assumed quasi-static during the sensing interval [32], [33]. In the m -th receiving antenna, the n -th received sample, $r_m[n]$, includes faded sample of signal, $s_m[n] = h_m x[n]$ with additive noise, $w_m[n]$. On the other hand, in the null hypothesis, H_0 , the sensed channel is idle, and the received samples only contain noise samples. Thus, the spectrum sensing problem can be expressed through the following binary hypothesis testing (BHT) problem

$$\begin{cases} H_0 : r_m[n] = w_m[n] \\ H_1 : r_m[n] = s_m[n] + w_m[n], \end{cases} \quad (2)$$

for $1 \leq m \leq M$ and $1 \leq n \leq N$. In this BHT problem, $s_m[n]$ is a complex unknown deterministic parameter, while $r_m[n]$ and $w_m[n]$ are complex random variables (RVs) following the same distribution with different locations. Moreover, we assume that samples of noise are independently and identically distributed (i.i.d.), and $s_m[n]$ and $w_m[n]$ are also independent. Furthermore, it should be mentioned that in this paper, noise samples follow isotropic distribution [34]. Consequently, in-phase (I) component of noise samples, $w_m^I[n]$, and quadrature (Q) component of noise samples, $w_m^Q[n]$, are statistically dependent. Therefore, especially under impulsive scenarios, the joint distribution of these components in the following section are presented where the distribution of each component and marginal distributions can be found numerically and with non-closed-form formula by integrating over the joint distribution. However, for the special case of complex Gaussian noise, the bivariate distribution is isotropic, and I and Q components of noise are statistically independent [34]. For noise samples, three different general non-Gaussian distributions based on literature in SS are employed: (i) MCA distribution, (ii) isotropic complex S α S distribution, and (iii) isotropic complex general Gaussian distribution (CGGD). In the following, these distributions are reviewed and the corresponding LRT for each one is obtained.

III. LIKELIHOOD RATIO TEST

In this paper, we firstly investigate the LRT for SS modeled by the BHT problem with the three mentioned noise distributions. By rewriting (2) in matrix forms with size $M \times N$, we have

$$\begin{cases} H_0 : \mathbf{R} = \mathbf{W} \\ H_1 : \mathbf{R} = \mathbf{S} + \mathbf{W}, \end{cases} \quad (3)$$

where the element in the m -th row and n -th column of $\mathbf{R}$ is $[\mathbf{R}]_{m,n} = r_m[n]$, and similarly $[\mathbf{S}]_{m,n} = s_m[n]$, $[\mathbf{W}]_{m,n} = w_m[n]$. Now, since only information about $\mathbf{S}$ is unavailable in the SU, the LRT for (3) is given by [35]

$$T_{LRT}(\mathbf{R}) = \max_{\mathbf{S}} L(\mathbf{R}|\mathbf{H}_1) - L(\mathbf{R}|\mathbf{H}_0) \geq \eta, \quad (4)$$

where $L(\mathbf{R}|\mathbf{H}_i)$ is the log-likelihood function (LLF) under $\mathbf{H}_i$. If the test statistic derived by LRT, $T_{LRT}(\mathbf{R})$, is greater than a detection threshold, η , $\mathbf{H}_1$ is confirmed, and on the other hand, $\mathbf{H}_0$ is considered as the condition of the sensed channel. It should be noticed that LRT has been the dominant solution for BHT problems not only in the SS context but also in a wide variety of fields related to BHT problems.

For comparing the performance of test statistics, probability of detection can be found by $P_d = \Pr[T(\mathbf{R}) > \eta|\mathbf{H}_1]$. Additionally, in order to lose just a few number of vacant channels due to false alarms by test statistic, probability of false alarm, $P_{fa} = \Pr[T(\mathbf{R}) > \eta|\mathbf{H}_0]$, should be less than or equal to a desired false alarm probability p_{fa} . To satisfy the constraint $P_{fa} \leq p_{fa}$, η should be appropriately selected before the sensing procedure based on distribution of $T(\mathbf{R})$ under $\mathbf{H}_0$ or by monte carlo (MC) simulations.

A. Noise Samples With Complex MCA Distribution

In the first case, we consider complex MCA distribution for $w_m[n]$, which is able to describe behavior of not only conventional thermal noise but also impulsive noise with heavier tails. Thus, noise samples contain two terms $w_m[n] = w_m^g[n] + w_m^{im}[n]$ where $w_m^g[n]$ models thermal or background noise, and the impulsive component, $w_m^{im}[n]$, includes other source of noise such as atmospheric and man-made noise or co-channel interference of a wireless network [4], [12]. Here, the number of identical noise sources emitting signal is denoted by P and is modeled by Poisson distribution ($P \sim \mathcal{P}(A)$). Additionally, noise signal of each source follows $\mathcal{CN}(0, \sigma^2)$ which is a zero-mean complex normal noise with variance σ^2 . Thus, the probability density function (PDF) of $w_m[n]$ is

$$f(w_m[n]) = \sum_{p=0}^{\infty} \Pr[P = p] \mathcal{CN}(0, \sigma_p^2), \quad (5)$$

where $\sigma_p^2 = \sigma_g^2 + p\sigma^2$, and σ_g^2 is the variance of background noise. The mean number of impulsive noise sources is $E\{P\} = A$ which is called impulsive index. By denoting variance of $w_m^{im}[n]$ as σ_{im}^2 and defining $\Xi = \sigma_g^2/\sigma_{im}^2$, the complex PDF of $w_m[n] \sim \mathcal{MCA}(A, \Xi, \sigma_w^2, \mu)$ can be written as follows

$$f(w_m[n]) = \sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p! \pi \sigma_p^2} \exp\left(-\frac{|w_m[n] - \mu|^2}{\sigma_p^2}\right), \quad (6)$$

where $\sigma_w^2 = \sigma_g^2 + \sigma_{im}^2$, and $\sigma_p^2 = \sigma_w^2(A^{-1}p + \Xi)/(1 + \Xi)$. Also, location parameter, μ , is equal to zero for noise samples.

By modeling the noise distribution of BHT problem in (2) with MCA distribution, we are able to write the distribution of the received signal under both hypotheses

$$\begin{cases} H_0 : r_m[n] \sim \mathcal{MCA}(A, \Xi, \sigma_w^2, 0) \\ H_1 : r_m[n] \sim \mathcal{MCA}(A, \Xi, \sigma_w^2, s_m[n]), \end{cases} \quad (7)$$

for $1 \leq m \leq M$ and $1 \leq n \leq N$. In this BHT problem, the distribution of noise samples can be fully determined in the receiver if all four parameters are available in the SU. Here, we assume all these parameters are known except for

$s_m[n]$, which should be estimated under H_1 . Therefore, since the noise samples are i.i.d., LLF of $\mathbf{R}$ under H_0 is

$$L(\mathbf{R}|H_0) = \sum_{m,n} \log \left[\sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p! \pi \sigma_p^2} \exp \left(-\frac{|r_m[n]|^2}{\sigma_p^2} \right) \right]. \quad (8)$$

For H_1 , we have

$$L(\mathbf{R}|H_1) = \sum_{m,n} \log \left[\sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p! \pi \sigma_p^2} \exp \left(-\frac{|r_m[n] - s_m[n]|^2}{\sigma_p^2} \right) \right], \quad (9)$$

where $\mathbf{S}$ is a deterministic unknown parameter. To find LRT, we need to solve the following optimization problem

$$\max_{\mathbf{S}} L(\mathbf{R}|H_1). \quad (10)$$

The optimum value for $\mathbf{S}$ is called maximum likelihood estimation (MLE). Here, we show the MLE of $\mathbf{S}$ by $\hat{\mathbf{S}}$. By putting $\hat{\mathbf{S}}$ in (10), the maximum value of cost function can be computed and plugged into (4) to find the LRT for (7). In the following proposition, MLE of (10) is obtained.

Proposition 1: For BHT in (7), we have

$$\hat{\mathbf{S}} = \mathbf{R} = \arg \max_{\mathbf{S}} L(\mathbf{R}|H_1). \quad (11)$$

Proof: See Appendix A. ■

Now, by use of $\hat{\mathbf{S}} = \mathbf{R}$, the optimum value of cost function in problem (10) becomes

$$\max_{\mathbf{S}} L(\mathbf{R}|H_1) = \sum_{m,n} \log \left[\sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p! \pi \sigma_p^2} \right] = c_0, \quad (12)$$

which is independent of the received matrix $\mathbf{R}$. Thus, we consider it as a constant, denoted by c_0 . The LRT formula in (4) for MCA noise model by use of (8) and (12) is equal to

$$T_{LRT-MCA}(\mathbf{R}) = c_0 - \sum_{m,n} \log \left[\sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p! \pi \sigma_p^2} \exp \left(-\frac{|r_m[n]|^2}{\sigma_p^2} \right) \right] \underset{H_0}{\overset{H_1}{\geq}} \eta. \quad (13)$$

For simulating this test, we truncate the inner summation with sufficiently large P_{end} as its upper bound.

B. Noise Samples With Isotropic Complex S α S Distribution

The main property of RVs from stable distribution is that the linear combination of independent stable RVs follows stable distribution as well. As a consequence, this property leads to a specific characteristic function (CF) for the real stable RVs. For instance, X with stable distribution, $X \sim \mathcal{S}(\alpha, \beta, \gamma_s, \mu)$, has the following CF [34], [36]

$$\begin{aligned} \Phi_X(u) &= E\{\exp(iuX)\} = \exp(i\mu u - \gamma_s^\alpha |u|^\alpha [1 - i\beta \kappa(u, \alpha)]), \\ \kappa(u, \alpha) &= \begin{cases} \text{sgn}(u) \tan \frac{\pi \alpha}{2}, & \alpha \neq 1 \\ -\frac{2}{\pi} \text{sgn}(u) \log(|u|), & \alpha = 1, \end{cases} \\ 0 < \alpha \leq 2, \quad -1 \leq \beta \leq 1, \quad 0 < \gamma_s. \end{aligned} \quad (14)$$

If $\beta = 0$, the distribution becomes symmetric around location parameter, μ , and is called symmetric α -stable (S α S). Therefore, for the real RV like X , the CF is

$$\Phi_X(u) = E\{\exp(iuX)\} = \exp(i\mu u - \gamma_s^\alpha |u|^\alpha), \quad (15)$$

where $u \in \mathbb{R}$, and γ_s is the scale parameter, and α is the shape parameter which changes the heaviness of the tails of S α S distribution.

A complex distribution of $X = X^I + iX^Q$ is isotropic (rotationally invariant) when the distribution of X and distribution of its rotated version $e^{i\theta} X$ (θ is an arbitrary angle of rotation) are identical [34]. Thus, X with complex S α S distribution, $X \sim \mathcal{S}\alpha\mathcal{S}(\alpha, \gamma, \mu)$, has the following CF [36]

$$\Phi_X(u) = E\{\exp(i\text{Re}[u^H X])\} = \exp(i\text{Re}[u^H \mu] - \gamma_s^\alpha |u|^\alpha), \quad (16)$$

where these scalars $X, u, \mu \in \mathbb{C}$, and superscript H shows hermitian operation. Now, for $w_m[n] \sim \mathcal{S}\alpha\mathcal{S}(\alpha, \gamma_s, 0)$, the CF is

$$\begin{aligned} \Phi_{w_m[n]}(u_{m,n}) &= E\{\exp(i\text{Re}[u_{m,n}^H w_m[n]])\} = \exp(-\gamma_s^\alpha |u_{m,n}|^\alpha), \end{aligned} \quad (17)$$

where $w_m[n] = w_m^I[n] + iw_m^Q[n]$, and $u_{m,n} = u_{m,n}^I + iu_{m,n}^Q$. It can be simply shown that for $e^{i\theta} w_m[n]$, its CF is $\Phi_{e^{i\theta} w_m[n]}(u_{m,n}) = \Phi_{w_m[n]}(u_{m,n})$. Thus, $w_m[n]$ with CF in (16) is isotropic. Now, we can rewrite BHT in (2) with use of isotropic complex S α S distribution of $w_m[n]$

$$\begin{cases} H_0 : r_m[n] \sim \mathcal{S}\alpha\mathcal{S}(\alpha, \gamma_s, 0) \\ H_1 : r_m[n] \sim \mathcal{S}\alpha\mathcal{S}(\alpha, \gamma_s, s_m[n]), \end{cases} \quad (18)$$

for $1 \leq m \leq M$ and $1 \leq n \leq N$, where $s_m[n]$ is a deterministic complex unknown parameter that just shifts the distribution under H_1 . Moreover, α and γ_s are considered as known parameters in this BHT problem. Since noise samples are assumed i.i.d., the LLF of the received matrix $\mathbf{R}$ under H_i is

$$L(\mathbf{R}|H_i) = \sum_{m,n} \log [\mathcal{F}^{-1}\{\Phi_{r_m[n]|H_i}(-u_{m,n})\}], \quad (19)$$

where $\mathcal{F}^{-1}\{\cdot\}$ is the inverse Fourier transform. Thus, the LRT by use of (4) and (16) is

$$\begin{aligned} T_{LRT}(\mathbf{R}) &= \max_{\mathbf{S}} L(\mathbf{R}|H_1) - \sum_{m,n} \log [\mathcal{F}^{-1}\{\exp(-\gamma_s^\alpha |u_{m,n}|^\alpha)\}]. \end{aligned} \quad (20)$$

To simplify (20), we require finding the maximum likelihood estimation (MLE) of $\mathbf{S}$ under H_1 denoted by $\hat{\mathbf{S}}$.

Proposition 2: For BHT in (18),

$$\hat{\mathbf{S}} = \mathbf{R} = \arg \max_{\mathbf{S}} L(\mathbf{R}|H_1). \quad (21)$$

Proof: See Appendix B. ■

Now, by using (16), (19), and (21), we have

$$\begin{aligned} \max_{\mathbf{S}} L(\mathbf{R}|H_1) &= \sum_{m,n} \log [\mathcal{F}^{-1}\{\exp(i\text{Re}(-u_{m,n}^H r_m[n]) - \gamma_s^\alpha |u_{m,n}|^\alpha)\}]. \end{aligned} \quad (22)$$

In the next step, for more simplification, we can use the 2-dimensional inverse Fourier transform formula which is $g(x_1, x_2) = \mathcal{F}^{-1}\{G(u_1, u_2)\} = c_f \iint G(u_1, u_2) e^{ix_1 u_1 + ix_2 u_2} du_1 du_2$. Thus,

$$\begin{aligned} \max_{\mathbf{S}} L(\mathbf{R}|H_1) &= \sum_{m,n} \log \left[c_f \iint \exp(-\gamma_s^\alpha |u_{m,n}|^\alpha) du_{m,n}^I du_{m,n}^Q \right]. \end{aligned} \quad (23)$$

Since (23) is not related to the received matrix $\mathbf{R}$, it equates to a constant, denoted by c_1 . Consequently, the LRT in (20) reduces to

$$T_{LRT}(\mathbf{R}) = c_1 - \sum_{m,n} \log [V(r_m[n], u_{m,n})], \quad (24)$$

where

$$\begin{aligned} V(r_m[n], u_{m,n}) &= c_f \left[\iint e^{i(u_{m,n}^I r_m^I[n] + u_{m,n}^Q r_m^Q[n]) - \gamma_s^\alpha [(u_{m,n}^I)^2 + (u_{m,n}^Q)^2]^{\alpha/2}} \right. \\ &\quad \left. du_{m,n}^I du_{m,n}^Q \right]. \end{aligned} \quad (25)$$

This LRT in (24) has no mathematical closed-form. However, for $\alpha = 2$, corresponding to complex Gaussian noise, it is equal to energy detector (ED) [15], [35]

$$T_{LRT}(\mathbf{R}) = \|\mathbf{R}\|_F^2, \quad (26)$$

where $\|\cdot\|_F$ is the Frobenius norm, and for $\alpha = 1$, S α S model is identical to isotropic complex Cauchy noise, and (24) reduces to Cauchy detector (CD) [15]

$$T_{LRT}(\mathbf{R}) = \sum_{m,n} \log(\gamma_s^2 + |r_m[n]|^2). \quad (27)$$

For other amounts of α , we compute (24) by approximation and discretized version of integral in (25)

$$\begin{aligned} \overline{V}(r_m[n], u_{m,n}) &= c_f \sum_{l=-L_s}^{L_s} \sum_{k=-K_s}^{K_s} V_{l,k}(r_m[n]) \Delta u^I \Delta u^Q \\ V_{l,k}(r_m[n]) &= e^{i(l\Delta u^I r_m^I[n] + k\Delta u^Q r_m^Q[n]) - \gamma_s^\alpha [(l\Delta u^I)^2 + (k\Delta u^Q)^2]^{\alpha/2}}, \end{aligned} \quad (28)$$

where we sampled $u_{m,n}^I$ and $u_{m,n}^Q$ with identical sampling interval for various amounts of n and m denoted by Δu^I and Δu^Q . Therefore, in the following, LRT can be found numerically by plugging (28) into (24). In the following, for Gaussian case with $\alpha = 2$, we validate this approximation by comparing results of LRT computed by (24) and (28) with ED in (26).

C. Noise Samples With Isotropic Complex Generalized Gaussian Distribution

Another complex non-Gaussian distribution for modeling noise samples is CGGD. If $X \in \mathbb{C}$ and $X \sim \mathcal{GG}(s, \gamma_g, \mu)$, the probability density function (PDF) of X is [5]

$$f_X(x) = \frac{s}{\pi \gamma_g \Gamma(1/s)} \exp(-|\frac{x - \mu}{\gamma_g}|^{2s}), \quad (29)$$

where s , γ_g , and μ are the shape, scale, and location parameters of the distribution, respectively, and $\Gamma(\cdot)$ denotes gamma function. For $s = 1$, CGGD is equal to the well-known complex Gaussian noise.

Here, in this BHT problem, we assume shape and scale parameters are known in the receiver of the SU. Thus, if the noise samples are distributed by CGGD, and samples of $s_m[n]$ are unknown parameters, then (2) becomes

$$\begin{cases} H_0 : r_m[n] \sim \mathcal{GG}(s, \gamma_g, 0) \\ H_1 : r_m[n] \sim \mathcal{GG}(s, \gamma_g, s_m[n]). \end{cases} \quad (30)$$

Now, by assuming i.i.d. noise samples for all antennas, the LLF function for $\mathbf{R}$ under H_1 is

$$L(\mathbf{R}|H_1) = MN \log \left(\frac{s}{\pi \gamma_g \Gamma(1/s)} \right) - \sum_{m,n} \left| \frac{r_m[n] - s_m[n]}{\gamma_g} \right|^{2s}, \quad (31)$$

while for H_0 , LLF has the same form only with $s_m[n] = 0$. To find the LRT for BHT problem in (30), MLE of $\mathbf{S}$ under H_1 should be computed. It is straightforward to prove that *Proposition 1* also holds for BHT problem in (30). Therefore, we have

$$\max_{\mathbf{S}} L(\mathbf{R}|H_1) = MN \log \left(\frac{s}{\pi \gamma_g \Gamma(1/s)} \right), \quad (32)$$

which is not related to the received signal. By plugging (31), and (32) into (4) and with a few manipulations, the LRT is

$$T_{LRT}(\mathbf{R}) = \sum_{m,n} \left| \frac{r_m[n]}{\gamma_g} \right|^{2s}. \quad (33)$$

This test is often referred to as fractional lower-order moments (FLOM) detector, and it is an effective test especially under noise with heavy-tailed distributions [18], [19].

IV. THE PROPOSED METHOD

A. Proposed Structure

In this section, we propose a detection method based on a special class of DNNs, which are called CNN. These types of DNNs are employed in various applications due to their satisfactory performance. Classification problems are also an important class of problems that can be handled by CNNs effectively.

The proposed CNN method includes two main parts; the first part is for feature extraction, and the next part of the network with fully connected (FC) layers is for partitioning. The first part is composed of convolutional and pooling layers. In convolutional layers, the signal is passed through a filter or mask with specific weights and size. In pooling layers, the

TABLE I
STRUCTURE AND LAYERS OF THE PROPOSED DETECTOR

Layer	Filters/Neurons	Window Size	Output Shape	AF
Input	–	–	$M \times N \times 2$	–
Conv2D	64	$1 \times 3 \times 2$	$M \times N \times 64$	ReLU
max pool	–	$1 \times 2 \times 1$	$M \times \frac{N}{2} \times 64$	–
Conv2D	64	$1 \times 3 \times 64$	$M \times \frac{N}{2} \times 64$	ReLU
max pool	–	$1 \times 2 \times 1$	$M \times \frac{N}{4} \times 64$	–
Conv2D	64	$1 \times 3 \times 64$	$M \times \frac{N}{4} \times 64$	ReLU
max pool	–	$1 \times 2 \times 1$	$M \times \frac{N}{8} \times 64$	–
Flattening	–	–	–	–
FC	128	–	–	ReLU
FC	64	–	–	ReLU
FC	1	–	–	Sigmoid

input of the layers is downsampled. Here, we employed max pooling layers which selects the sample with the maximum amount in a window. Sufficient number of kernels and proper size of kernel are selected for these layers of the proposed network to allow them to effectively extract frequent patterns embedded in the received signal, which are useful in classification of the received signal into two hypotheses of SS problem. In addition to extracting frequent patterns, these layers remove the infrequent patterns which are probably added to the main structure of signal in each hypothesis because of noise signal. Moreover, these layers downsample the received signal into effective features and help the partitioning part to classify the received signal with a smaller extent of complexity. The architecture of the proposed CNN for SS is shown in Table I.

In the feature extraction part, three 2-dimensional convolutional layers (Conv2D) are employed with rectified linear unit (ReLU) activation function (AF). This AF can be formulated by $f(x) = \max\{0, x\}$. After each of these layers, max pooling layers diminish the dimensions of the feature space. The following three FC layers can classify the signal into two hypotheses. The last FC layer is a single neuron with Sigmoid AF which is $f(x) = (1 + \exp(-x))^{-1}$, while neurons in the previous layers have ReLU AF. Furthermore, the original baseband samples of I and Q components of the received signal in various antennas ($r_m^I[n], r_m^Q[n]$) are stacked in an $M \times N \times 2$ array as input to the network.

The structure of the proposed CNN is selected based on various simulations in order to perform adequately for the SS problem modeled by BHT of (2) under general noise distributions. Here, we enumerate plausible reasons behind this selection. Firstly, in contrast to the majority of DNNs in the literature, the original samples of the received signals are fed to the network instead of SOMs for having the ability to handle impulsive models. Secondly, several convolutional and pooling layers extract related features and offer the ability of generalization for a variety of families of distributions

(MCA, CGGD, and S α S), different noise distributions within a family (from Gaussian to severely heavy-tailed distributions created by different statistical parameters), and also test signals with distinctive amount of power in comparison to power of training signal. Thirdly, the selected architecture has sufficient capacity to tackle complex SS problems, and concurrently, with use of dropout technique, it maintains a proper balance between overfitting and underfitting aiming to simultaneously provide high detection probability with generalization ability. Section V indicates the promising performance of the proposed CNN in comparison with other methods.

B. Learning Stage

The proposed CNN includes various types of layers, and weights in filters of Conv2D and FC layers which should be determined through the learning stage. To find the weights of these layers, an optimization problem has to be solved with appropriate cost function. Thus, here, we introduce a well-known cost function in training of DNNs, called cross-entropy (CE) [37], to form the optimization problem. A training dataset provides a posteriori probability $p_d(H_i|\mathbf{R})$, which indicates the allocated chance of belonging the received matrix of data $\mathbf{R}$ to hypothesis H_i by the dataset. For instance, if the k -th data matrix sample, $\mathbf{R}^{(k)}$, is labeled H_0 by dataset, then we can infer $p_d(H_0|\mathbf{R}^{(k)}) = 1$ and $p_d(H_1|\mathbf{R}^{(k)}) = 0$. If signal $\mathbf{R}$ does not belong to the training dataset, $p_d(H_i|\mathbf{R}) = 0$ for $i = 0, 1$. Hence, the probability offered by a dataset is non-zero just for the finite number of signals belonging to the training dataset. Now, we are able to write the CE cost function [37]

$$C(\mathbf{W}^{(L)}) = -\mathbb{E}_{\text{data}}\{\log(p_{\text{net}}(H_i|\mathbf{R}, \mathbf{W}^{(L)}))\}, \quad (34)$$

where $\mathbf{W}^{(L)}$ contains all the weights of the CNN from the first layer to the last layer denoted by L . Also, $p_{\text{net}}(H_i|\mathbf{R}, \mathbf{W}^{(L)})$ specifies the predicted probability by the network with set of weights in $\mathbf{W}^{(L)}$ for the event that $\mathbf{R}$ belongs to H_i . In CE

cost function, $E_{\text{data}}\{\cdot\}$ denotes the expected value on the probability space offered by the training dataset. Thus, we can use $p_d(H_i, \mathbf{R}) = p_d(H_i|\mathbf{R})p(\mathbf{R})$ for computing the expected value in CE cost function.

Here, we explain that if the proposed CNN can be perfectly trained over a proper dataset with CE cost function, it would be the MAP detector of the training dataset and minimize the Bayesian error [35]. To show that, we write the output of the Sigmoid function in the last neuron of the proposed CNN as

$$f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)}) \underset{H_0}{\overset{H_1}{\geq}} 0.5, \quad (35)$$

where it is a function of the input matrix data and set of all weights in $\mathbf{W}^{(L)}$. Also, Sigmoid is the AF of the single neuron in the last layer of the network and is denoted by $f^{(L)}$. Since this AF is in the range of $[0, 1]$, its output offers the predicted probability by the network for the event that the input signal belongs to H_1 , and we can write $p_{\text{net}}(H_1|\mathbf{R}, \mathbf{W}^{(L)}) = f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)})$. Therefore, if this probability is over 0.5, H_1 is the predicted class, or vice versa. Now, by rewriting (34), we have

$$\begin{aligned} C(\mathbf{W}^{(L)}) &= - \sum_i \int_{\mathbf{R}} p_d(H_i|\mathbf{R})p(\mathbf{R}) \log(p_{\text{net}}(H_i|\mathbf{R}, \mathbf{W}^{(L)})) d\mathbf{R}. \end{aligned} \quad (36)$$

With more simplification, it becomes

$$\begin{aligned} C(\mathbf{W}^{(L)}) &= - \int_{\mathbf{R}} \left[(1 - p_d(H_1|\mathbf{R}))p(\mathbf{R}) \log(1 - f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)})) \right. \\ &\quad \left. + p_d(H_1|\mathbf{R})p(\mathbf{R}) \log(f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)})) \right] d\mathbf{R}, \end{aligned} \quad (37)$$

where we use the fact that $p_d(H_0|\mathbf{R}) = 1 - p_d(H_1|\mathbf{R})$. This function is convex with respect to $f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)})$. Thus, if we assume that $\mathbf{W}^{(L)*}$ is the optimum value that results in the minimum of CE cost function, $C(\mathbf{W}^{(L)*})$, then by equating $\partial C(\mathbf{W}^{(L)*})/\partial f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)*})$ to zero, we can conclude that

$$\frac{1 - p_d(H_1|\mathbf{R})}{1 - f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)*})} = \frac{p_d(H_1|\mathbf{R})}{f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)*})}, \quad (38)$$

and with simple manipulations, we reach to

$$f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)*}) = p_d(H_1|\mathbf{R}). \quad (39)$$

Also, in this ideal case, by plugging $\mathbf{W}^{(L)*}$ into (35) and use of (39), it implies that the decision rule for the predicted class by the proposed CNN with optimum weights is based on

$$f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)*}) \underset{H_0}{\overset{H_1}{\geq}} p_d(H_1|\mathbf{R}). \quad (40)$$

Therefore, if we can ideally train the network to find the optimum weights ($\mathbf{W}^{(L)*}$), which minimizes CE cost function, then the trained network is the MAP detector of the training dataset.

Algorithm 1 Generating Complex MCA RV

Require: $0 < A$, $0 < \Xi$, $0 < \sigma_w^2$

generate $p \sim \mathcal{P}(A)$
 $\sigma_p^2 \leftarrow \sigma_w^2 \frac{pA^{-1} + \Xi}{1 + \Xi}$
 generate $g^i, g^q \sim \mathcal{N}(0, 1)$
 $w \leftarrow (\sigma_p^2/2)^{1/2}(g^i + ig^q)$
return w

It should be noted that $\mathbf{W}^{(L)*}$ is the optimum set of weights, and it is hard to be found for the problem at hand. However, by taking appropriate measures, we can find a proper set of weights of the proposed CNN for the problem. Thus, besides selecting a suitable structure for DNN, training the network with an adequate number of epochs is highly effective to reach a set of near-optimum weights after the learning stage. In the next section, we discuss finding an adequate number of epochs for training the network.

Moreover, by knowing p_{fa} , we change the bias of the last neuron to satisfy $P_{fa} = p_{fa}$. For further explanation, we write $f^{(L)}(\mathbf{R}, \mathbf{W}^{(L)}) = \text{Sigmoid}(\mathbf{w}^{(L)T} f^{(L-1)}(\mathbf{R}, \mathbf{W}^{(L-1)}) + b^{(L)})$ where $\mathbf{w}^{(L)}$ is the vector of the weights from the output vector of the $(L-1)$ -th layer ($f^{(L-1)}(\mathbf{R}, \mathbf{W}^{(L-1)})$) to the L -th layer, and $b^{(L)}$ is the bias of the single neuron in the last layer. Now, we can rewrite (35) as $\mathbf{w}^{(L)T} f^{(L-1)}(\mathbf{R}, \mathbf{W}^{(L-1)}) > -b^{(L)}$ for confirming H_1 . Since it is hard to find analytic relationship between P_{fa} and $b^{(L)}$ for CNN, here, we simply use decision of CNN with various $b^{(L)}$ for K signals from H_0 . Then, based on these decisions and by means of indicator function $\mathbb{I}\{\cdot\}$, we compute P_{fa} for different $b^{(L)}$, and we select it in order to satisfy $P_{fa} = \frac{1}{K} \sum_k \mathbb{I}\{\mathbf{w}^{(L)T} f^{(L-1)}(\mathbf{R}^{(k)}, \mathbf{W}^{(L-1)}) > -b^{(L)} | H_0\} \simeq p_{fa}$. Clearly, computation of P_{fa} becomes more accurate by increasing K .

V. SIMULATION RESULTS

In this part, we present the performance of the proposed CNN in comparison with the particular derived LRT solution for each of the three signal models and other state-of-the-art methods. For the setup of simulations, we mainly consider $N = 20$, $M = 4$, and $p_{fa} = 0.1$, and generation of 80000 signals for training dataset with the same number of signals from both classes.

To clarify noise models, in Algorithm 1, pseudocode of MCA model employed for related simulations is presented. We should also mention that for this noise model, the signal-to-noise ratio (SNR) is $\text{SNR} = P_0/\sigma_w^2$ where $E\{|x[n]|^2\} = P_0$. Next, we use gamma distribution (denoted by $\mathcal{G}$) and uniform distribution (denoted by $\mathcal{U}$) for generating RVs with isometric CGGD (see [5]). The details of this implementation are explained in Algorithm 2. Since the variance of the generated final RV (denoted by w) in Algorithm 2 is $\gamma_g^2 \Gamma(2/s)/\Gamma(1/s)$ [5], we use $\text{SNR} = P_0/(\gamma_g^2 \Gamma(2/s)/\Gamma(1/s))$ as SNR formula in problems with isometric CGGD. Finally, Algorithm 3 states realization of isometric complex bivariate SαS compactly [34], [36]. In this algorithm, we use the fact that for Gaussian case, the isometric

Algorithm 2 Generating Isometric Complex Generalized Gaussian RV

Require: $0 < s, 0 < \gamma_g$
generate $z_0 \sim \mathcal{G}(1/s, 1)$
generate $z_1 \sim \mathcal{U}(0, 1)$
 $w \leftarrow \gamma_g z_0^{1/2s} \exp\{2\pi i z_1\}$
return w

Algorithm 3 Generating Isometric Complex S α S RV

Require: $0 < \alpha \leq 2, 0 < \gamma_s, \mu \in \mathbb{C}$
if $\alpha = 2$ **then**
generate $w^i \sim \mathcal{S}\alpha\mathcal{S}(\alpha, \gamma_s, \text{Re}\{\mu\})$
generate $w^q \sim \mathcal{S}\alpha\mathcal{S}(\alpha, \gamma_s, \text{Im}\{\mu\})$
 $w \leftarrow w^i + iw^q$
return w
else
 $\gamma_1 \leftarrow 2\gamma_s \cos(\pi\alpha/4)^{\frac{2}{\alpha}}$
generate $z \sim \mathcal{S}(\alpha/2, 1, \gamma_1, 0)$
generate $g^i, g^q \sim \mathcal{N}(0, 1)$
 $w \leftarrow z^{\frac{1}{\alpha}}(g^i + ig^q) + \mu$
return w
end if

distribution has independent I and Q components, while for $\alpha < 2$, the generated complex RV is only isometric [34]. Also, for this distribution, the transmitted power in PU is based on the generalized signal-to-noise ratio (GSNR) with this definition $\text{GSNR} = P_0/4\gamma_s$. GSNR is employed in this case since for $\alpha < 2$ the variance is undefined.

As already mentioned, we require selecting the optimum number of epochs for the training stage. Here, we adjusted the bias of the last layer of the proposed CNN according to the method explained in Section IV-B and [16]. Fig.1 includes the probability of detection of the proposed CNN for the number of epochs used in the training stage. For this part, the noise model follows the isometric complex S α S distribution with $\alpha = 2$ and $\gamma_s = 1$, which reduces to conventional complex Gaussian noise (CGGD becomes complex Gaussian for $s = 1$, and MCA also can approximately be considered as Gaussian distribution in cases with specific parameters as well). Each curve is corresponding to a test set with a special GSNR which is used for computing the probability of detection. It is clear that in this case, 15 epochs of training offer the best probability of detection. Thus, the same method can also improve the performance of DNN-based detectors and is employed in the following to select the appropriate number of training epochs.

In the following comparative simulations, we also compute the probability of detection of several detectors in the literature. In this part, the well-known ED, CD, and COS detector of the first and M -th antenna proposed in [15] are simulated. In addition to simulation of model-based detectors like CCAV [9], PCA [21], and scaled largest eigenvalue (SLE) [32], [39], these figures include results of data-driven detectors such as ANN in [23] and SAE in [30]. Additionally, the detection probability of derived LRTs for each specific case is presented to compare the exclusively derived LRT of

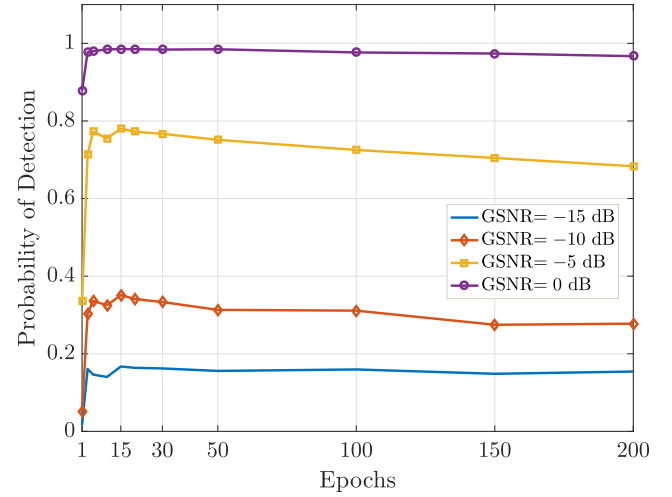


Fig. 1. Probability of detection of the proposed CNN versus the number of the training epochs for various GSNRs (dB) with S α S and $\alpha = 2$ and $\gamma_s = 1$.

each noise model with the proposed CNN. It is noteworthy to mention that although FLOM detector is LRT of CGGD, its performance is provided in other cases as well since it is an effective detector under all cases of impulsive noise.

A. Probability of Detection

1) *MCA*: Here, we consider the MCA distribution and related SS problem stated in (7), and we find the detection probability of the proposed CNN and related LRT in (13) for this problem. It is worthwhile to explain that for simulating LRT_{MCA} in (13), the inner summation is truncated when p takes sufficiently and appropriately large values ($p = P_{end} = 100$). Also, other aforementioned detectors are included such as FLOM in (33) with $s = 0.01$. In Fig. 2, probability of detection for different amounts of SNR with these parameters of MCA noise model $A = 0.05, \Xi = 0.01$ and $\sigma_w^2 = 1$ are plotted for the mentioned detectors. As presented, for this impulsive noise model, the proposed CNN is the best detector while it is followed by FLOM and LRT_{MCA} . For instance, $P_d = 0.8$ is achieved by CNN with $\text{SNR} = -24.1$ dB while LRT_{MCA} and FLOM both require -20.8 dB. Other model-based detectors and data-driven detectors perform inadequately in this case (results of DNN-based detectors are shown by curves with star marker).

In the next scenario, the same curves are presented for $A = 0.005, \Xi = 0.1$, and $\sigma_w^2 = 1$ in Fig. 3. For this scenario, again, the proposed CNN achieves the highest probability of detection for various SNRs. Also, LRT_{MCA} , FLOM, and ANN have close but inferior performance to that of the proposed CNN. Therefore, we can conclude that the proposed CNN offers a satisfactory performance under MCA model compared to other detectors.

2) *Isometric Complex S α S*: For the SS problem in (18) with isometric complex S α S distribution, we can investigate the performance of the proposed CNN both under conventional Gaussian noise and impulsive noise. By setting $\alpha = 2$, and $\gamma_s = 1$, this distribution reduces to the conventional circularly

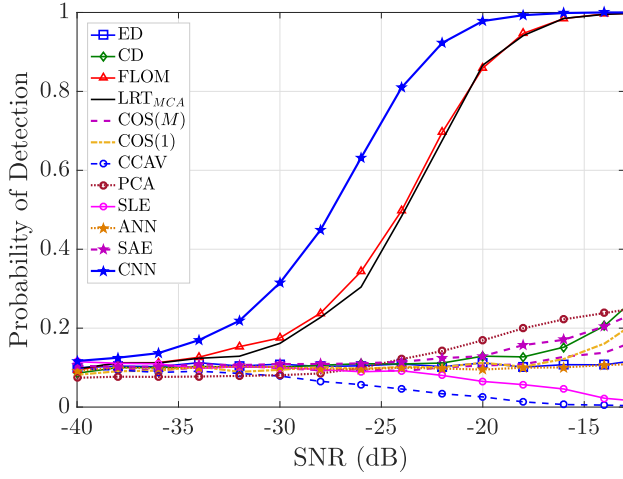


Fig. 2. Probability of detection of the detectors versus the SNR (dB) for the noise signal with MCA noise model and $A = 0.05$, $\Xi = 0.01$.

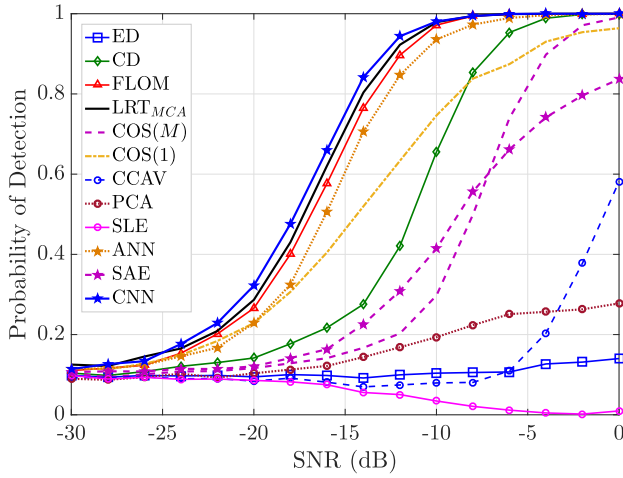


Fig. 3. Probability of detection of the detectors versus the SNR (dB) for the noise signal with MCA noise model and $A = 0.005$, $\Xi = 0.1$.

complex Gaussian distribution, and also LRT in (24) becomes equal to ED formula in (26). Additionally, we report the simulation results of LRT with approximation used in (28) by $LRT_{S\alpha S}$ where we set its parameters as $L_s = K_s = 16$ and $\Delta u^I = \Delta u^Q = 0.5$ for the rest of this section. In Fig. 4, ED and $LRT_{S\alpha S}$ have identical curves which validate the approximation in (28). As expected, for detection of the unstructured signal in Gaussian noise model like the problem of (18) with Gaussian noise, ED is the optimum detector [35], [40]. We can observe that the performance of the proposed CNN is very close to that of ED. Moreover, ANN and CD have a quite close performance to ED. In this case, ANN is expected to have a close performance to that of ED since it employs energy features of the received signal as input to the network.

Next, the isometric impulsive noise is regarded with $\alpha = 1.5$ and $\gamma_s = 1$ in Fig. 5. In this scenario, the proposed CNN outperforms other detectors. However, in contrast to the previous scenario, performance of ED and ANN degrades severely since the energy-based detectors are heavily susceptible to

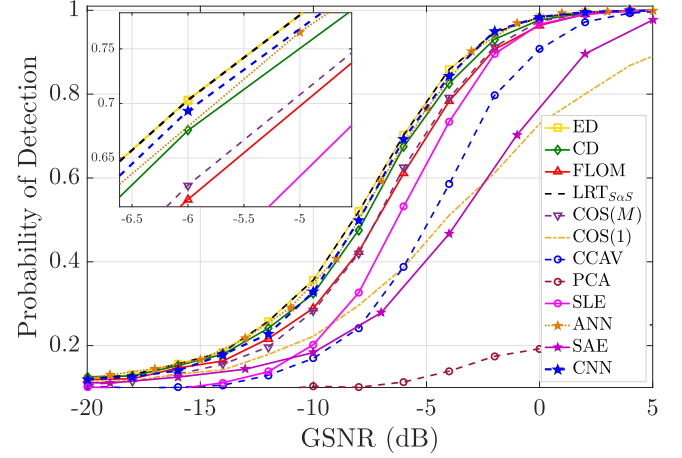


Fig. 4. Probability of detection of the detectors versus the GSNR (dB) for the noise signal with $S\alpha S$ noise model and $\alpha = 2$, $\gamma_s = 1$.

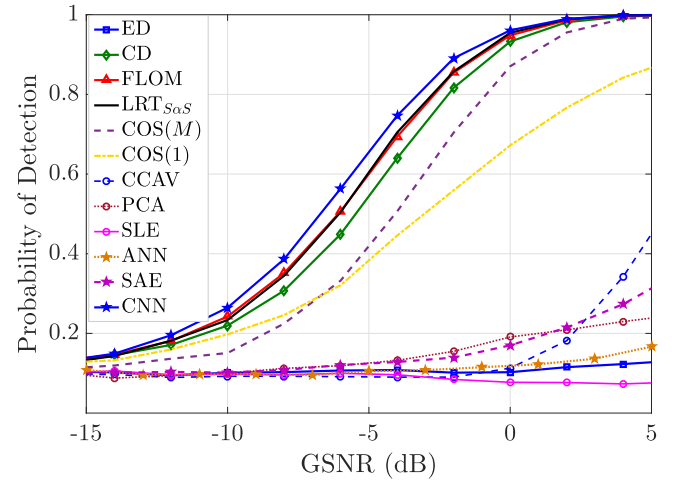


Fig. 5. Probability of detection of the detectors versus the GSNR (dB) for the noise signal with $S\alpha S$ noise model and $\alpha = 1.5$, $\gamma_s = 1$.

impulsive noise [15], [16]. After the CNN-based detector, $LRT_{S\alpha S}$, FLOM, and CD offer the highest detection probability, respectively.

By decreasing α to 1.2, impulses in noise are increased, which leads to a larger gap between detection probability curves of the proposed CNN and other detectors such as $LRT_{S\alpha S}$ and FLOM. This gap and the superiority of the proposed method are shown in Fig. 6. Again, under this noise model, we observed that the proposed CNN not only surpasses other detectors in cases with $\alpha < 2$ but also offers a performance close to the optimum performance of ED in Gaussian case. Moreover, the proposed data-driven method is not susceptible to change of models' parameters such as α in contrast to detectors like ED and ANN.

3) *Isometric CGGD*: Finally, we examine the isometric CGGD for noise model in SS problem similar to the problem in (30). For this noise model, the LRT derivation leads to FLOM as shown in Section III-C and (31). Thus, in these simulations, we use the true value of s in signal model for FLOM detector. Fig. 7 includes probability of predicting

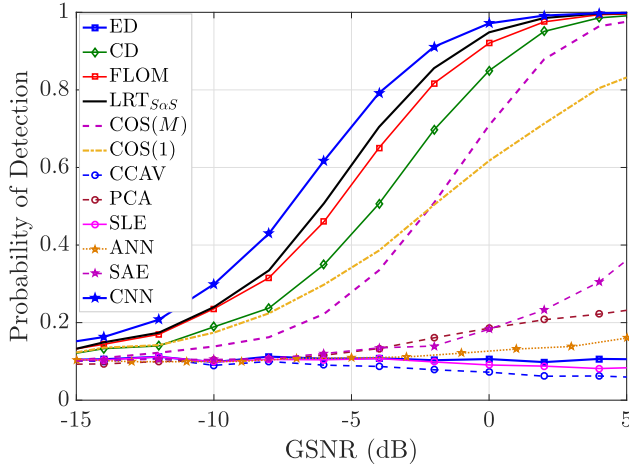


Fig. 6. Probability of detection of the detectors versus the GSNR (dB) for the noise signal with SαS noise model and $\alpha = 1.2$, $\gamma_s = 1$.

H_1 by detectors when the transmitted signal by PU is contaminated by isometric CGGD noise with $s = 0.2$ and $\gamma_g = 1$. Despite the previous noise models, CD has the second-highest correct detection probability after the proposed CNN while FLOM (LRT in this case) requires about 2 dB and 4 dB more SNR than CD and CNN for gaining $P_d = 0.8$. For the next set of parameters with $s = 0.4$, performance of all detectors is presented in Fig. 8. As seen, similar to several previous scenarios, detection probability of CNN exceeds that of LRT and CD in this scenario.

By regarding simulation results of all three models with various parameters, we can conclude notable points. Firstly, in all simulations under these three models, CNN's performance is the best, or it gains a P_d near to that of optimum detectors where this detector exists. This happens in the problem at hand when general noise models reduce to conventional Gaussian case such as SαS model with $\alpha = 2$ in Fig. 4. Similarly, for MCA with large Ξ and CGGD with $s = 1$, LRT and distributions turn into ED and complex Gaussian distribution. Thus, we exclude these cases since Fig. 4 provides the same condition and shows the near-optimum performance of the proposed CNN in these cases. This also verifies that with a sufficiently rich training dataset and appropriate number of training epochs, the obtained set of weights, $\mathbf{W}^{(L)}$, is close to $\mathbf{W}^{(L)*}$ which results in near-optimum performance of CNN. Next, in these three general noise models, CNN of this work, LRT, CD, and FLOM give the best detection probability. However, the proposed CNN has a robust performance for each of the models owing to its model-free property, while CD performs poorly with MCA model, and FLOM (LRT) has a degraded performance in the case of CGGD.

B. Discrepancy of Models or Datasets

One important issue raised in realistic scenarios is that an SU may use a detector adjusted for a particular environment with its specific signal model in another environment or in the same place but with different conditions. Since SU's position or condition of environment might alter, discrepancy

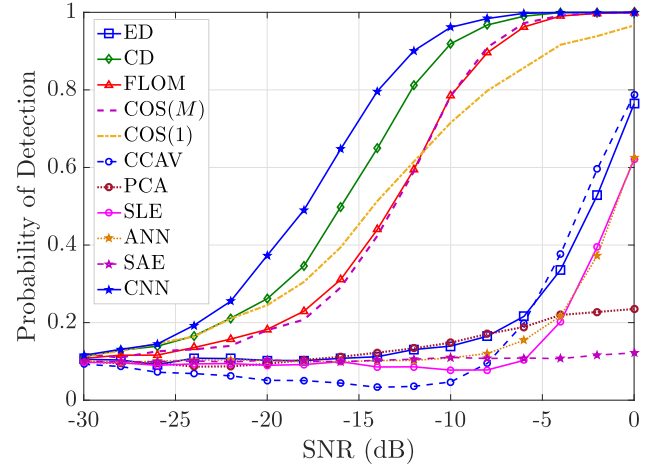


Fig. 7. Probability of detection of the detectors versus the SNR (dB) for the noise signal with CGGD and $s = 0.2$, $\gamma_g = 1$.

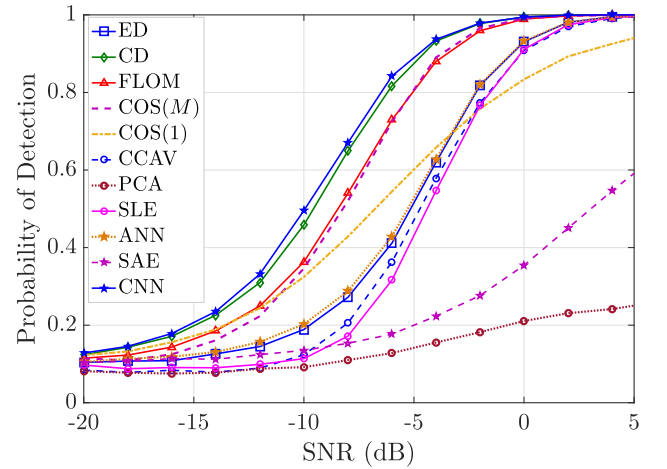


Fig. 8. Probability of detection of the detectors versus the SNR (dB) for the noise signal with CGGD and $s = 0.4$, $\gamma_g = 1$.

of models and datasets is a probable event that have a strong influence on the performance of detectors. Also, the same event may happen due to estimation error of model parameters. To take this influence into consideration, here, we will train DNN-based detectors and adjust the detection threshold (η) of model-dependent detectors with a set of parameters and a training dataset. However, we evaluate their probability of detection with a different set of parameters and a test dataset generated with different parameters.

In Fig. 9, the detection threshold and weights of detectors are set based on the generated train dataset with MCA model and with parameters of $(A, \Xi) = (0.1, 0.01)$. Furthermore, we increased the number of receiving antennas into $M = 6$ for investigating its effect on detectors. Then, we find P_d for detectors under perfect scenario (the same model and parameter of train dataset are used), case of uncertainty with 20%, and extremem case of 50% error in estimation of parameters. Thus, in the case of uncertainty, detection probability is computed on generated test datasets with $(A^{20\%}, \Xi^{20\%}) = (0.08, 0.008)$ and $(A^{50\%}, \Xi^{50\%}) = (0.05, 0.005)$. In this figure, we only

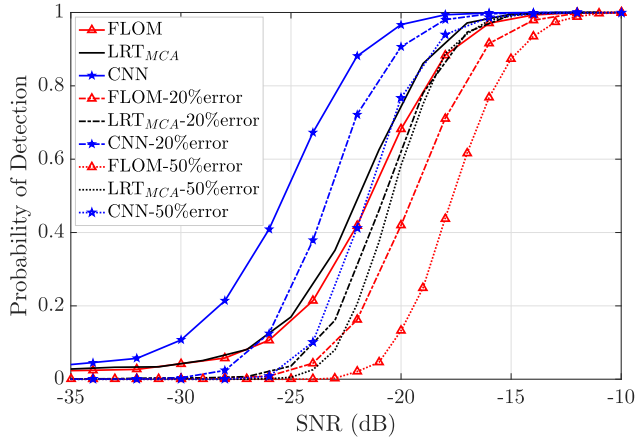


Fig. 9. Probability of detection of the detectors versus the SNR (dB) for the noise signal with MCA noise model, $M = 6$, and 20% and 50% error in true value of noise parameters.

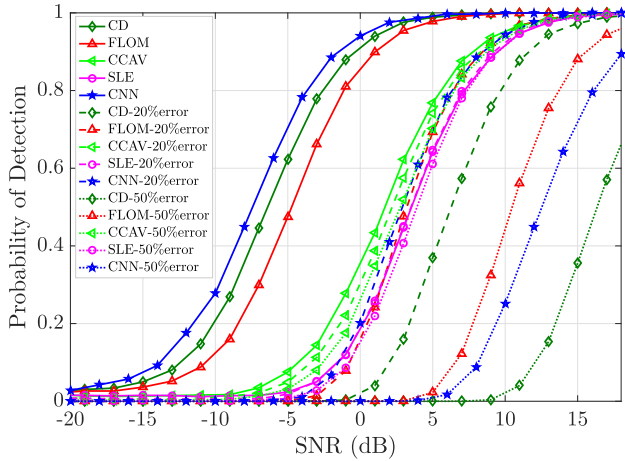


Fig. 10. Probability of detection of the detectors versus the SNR (dB) for the noise signal with CGGD noise model, $M = 2$, and 20% and 50% error in true value of noise parameters.

consider CNN, LRT_{MCA} , and FLOM due to their appropriate performance based on previous simulations. Whenever SU intends to act more opportunistically to lose fewer space holes, the p_{fa} should be decreased. Hence, in Fig. 9, the desired false alarm probability is $p_{fa} = 0.02$. The proposed CNN is the superior detector in the perfect match of models. For having $P_d = 0.6$, FLOM and LRT_{MCA} need about 3.8 dB and 3.3 dB greater SNR compared to CNN. Under both cases of mismatch of datasets (dotted curves), the proposed CNN performs better than LRT_{MCA} and FLOM as well. Even under 20% mismatch of datasets, CNN outperforms both other detectors in perfect scenarios for $SNR > -26$ dB.

Next, isometric CGGD is used as a model for producing signal for training and threshold adjustment with $(s, \gamma_g) = (0.3, 1)$. However, the same family of distribution with 20% difference of parameters, $(s^{20\%}, \gamma_g^{20\%}) = (0.36, 0.8)$ and 50% difference of parameters, $(s^{50\%}, \gamma_g^{50\%}) = (0.45, 0.5)$, are employed to evaluate detection probability. Curves of P_d versus SNR are shown for various detectors in Fig. 10 for CGGD with decreased number of antennas to $M = 2$ and

$p_{fa} = 0.02$. In this figure, the proposed CNN has the best performance for the error-free case, while its performance degrades under mismatch of parameters. CCAV and SLE show quite complete robustness against mismatch of parameters. This was expected since SLE has constant false alarm ratio (CFAR) property with respect to variance [39], [41]. However, since their performance in perfect scenario are largely inferior in comparison to CNN, their error-free and their dotted curves are also close to that of CNN and FLOM with 20% error. The extreme case of uncertainty with 50% error has severe degrading impact on the proposed CNN and FLOM while SLE and CCAV again have complete robust performance. Under this case, FLOM outperforms the proposed CNN. This indicates that under CGGD noise model with extreme case of uncertainty, compensating methods should be concerned for the proposed CNN. In these situations where the discrepancy between training and test datasets degrades the performance of DNNs, weights of the network can be updated by acquiring new signals in the new environment and by using these signals for a fine-tuning process. This transfer learning approach appeared to be highly promising in scenarios for symbol detection where perfect channel information is not available [42]. Thus, the same method can also be developed for SS under extreme case of discrepancy between the train and test datasets. Moreover, we can conclude that in the error-free case, CNN outperforms distinctively, and it can be inferred that for more moderate discrepancy of parameters or datasets, CNN has a better performance. Additionally, we observed that performance of SLE and CCAV in impulsive scenarios with MCA and S α S models are extremely inferior, which turns them unsuitable for SS with these models whether under error-free conditions or mismatch of parameters.

C. Comparison With Other DNNs

In addition to model-based detectors and those based on FC layers such as SAE and ANN, in this part, we compare the proposed CNN to the state-of-the-art RNN with LSTM cells [43], CM-CNN in [27], and FLOM-CNN in [31]. Moreover, we decreased the number of signals in training dataset to 40000 signals with just 5 different SNRs (GSNRs) shown by red spots in Fig. 11 and 12. However, the test datasets for finding P_d include signals with SNR (GSNR) which are new for the trained network. Therefore, the ability of networks to detect test signals with new amount of SNR (GSNR), which are not observed in the process of learning, can be evaluated [44].

In Fig. 11, the proposed CNN and mentioned DNNs are trained with signals contaminated by noise with S α S distribution, $\alpha = 1.2$, and $\gamma_s = 1$. The GSNR of these training signals are chosen from $GSNR_{tr} = \{-15, -10, -5, 0, 5\}$ dB while the interval of test GSNR, denoted by $GSNR_{te}$, has much wider range. The proposed CNN and LSTM outperform others, and they have close performance for $GSNR_{te} < 15$ dB. For larger $GSNR_{te}$, in contrast to the proposed CNN, LSTM is not able to generalize its effective performance for $GSNR_{te} > 15$ dB, and its performance degrades severely for large $GSNR_{te}$. CM-CNN's performance is poor due to

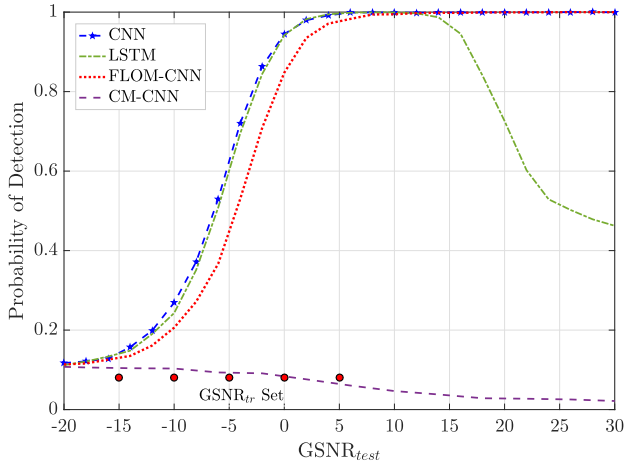


Fig. 11. Probability of detection of DNNs versus the GSNR (dB) for the noise signal with $S\alpha S$ distribution, $\alpha = 1.2$, $\gamma_s = 1$, and signals with different GSNR in training and test datasets.

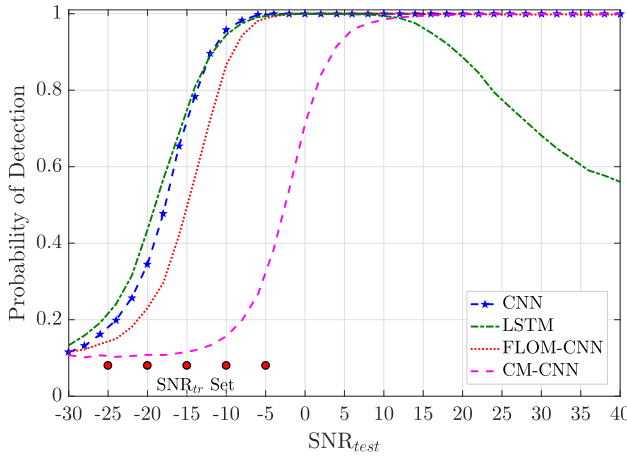


Fig. 12. Probability of detection of DNNs versus the SNR (dB) for the noise signal with CGGD distribution, $s = 0.2$, $\gamma_g = 1$, and signals with different SNR in training and test datasets.

using second-order moments (covariance matrix) under $S\alpha S$ distribution while FLOM-CNN can handle this noise model since its inputted matrix is computed based on FLOM.

The result of similar simulation with CGGD noise model, $s = 0.2$, $\gamma_g = 1$, and $SNR_{tr} = \{-25, -20, -15, -10, -5\}$ dB is illustrated in Fig. 12. Similar to the previous case, although LSTM slightly outperforms the proposed CNN for low SNR_{test} , it suffers from the same problem, and its probability of detection starts to degrade severely for $SNR_{te} > 10$ dB. It can be problematic for the SU with an LSTM network to detect a PU at short distance which probably has high SNR_{test} ($GSNR_{test}$). On the other hand, performance of the proposed CNN is robust for the entire interval of SNR_{te} . Moreover, it is worthwhile to compare Fig. 11 and 12 to Fig. 6 and 7 correspondingly, since the simulation settings are identical except for SNR (GSNR) of the training signals and the number of training signals which are doubled in Fig. 6 and 7. As seen, the proposed CNN is also able to generalize its robust performance over test signals with different power,

which are not observed in the learning stage, and even with a smaller training dataset.

D. Complexity

Finally, to compare detectors from other aspects, we use the number of required computational operations for a detector. The main burden of complexity can be considered as a function of the number of samples N and the number of receiving antennas M . Similar to [30], we focus on the number of required real multiplication as a function of M and N for complexity and the order of this function. Thus, we exclude other operations such as the number of logarithmic function, addition, AF in DNN-based detectors, and exponentiation (such as exponentiation with base $|r_m[n]|^2$ and exponent s in FLOM) except for squaring which is equal to one multiplication (four real multiplication for complex numbers).

It is easy to show that the number of multiplications for ED, CD, and FLOM is $2MN$. For LRT_{MCA} and $LRT_{S\alpha S}$, the dominant term in the number of real multiplications are equal to $2(P_{end} + 1)MN$ and $2(2L_s + 1)(2K_s + 1)MN$. For the proposed CNN, the exact number of real multiplications is $3MN \sum_{c_l=1}^{C_l} N_{f,c_l-1} N_{f,c_l} 2^{(1-c_l)} + \sum_{f_l=1}^{F_l} N_{n,f_l-1} N_{n,f_l}$. In this formula, C_l , F_l , N_{f,c_l} , and N_{n,f_l} are the number of convolutional layers, fully connected layers, input feature maps to c_l -th convolutional layer, and input features to f_l -th fully connected layer. Also, $N_{f,0} = 2$ due to two I and Q components of the received signal, and $N_{n,0}$ is the number of features in the output of the last convolutional layer ($M \times N/8 \times N_{f,C_l}$). For instance, with $M = 4$ and $N = 20$, the number of required multiplications for ED, CD, and FLOM is 160. LRT_{MCA} with truncated summation in $P_{end} = 100$, needs 16160 multiplications (dominant term). Also, the number of required multiplications (dominant term) for $LRT_{S\alpha S}$ with $L_s = K_s = 16$ is 174132 while for the proposed CNN with structure in Table I, it is 841792, for CM-CNN with input shape of (10, 10, 2), it is 826000, and for FLOM-CNN, it is 28145 with input shape of (10, 10, 2). Also, based on [30], the required number of multiplications for SAE with the same hyperparameters in [30] is 21050. This shows the higher complexity of the proposed CNN in comparison to other model-based and DNN-based detectors. In [16], a simple method for decreasing the complexity of CNN-based detectors by decreasing the number of convolutional filters has been proposed, which can be similarly used here.

VI. CONCLUSION

In this work, we investigated the SS problem with different general noise distributions such as MCA, isometric complex $S\alpha S$, and isometric CGGD. Since these distributions can model the behavior of noise in diverse environments, our proposed SS scheme is quite versatile. Our scheme is a CNN-based detector for BHT with these distributions. We verified that our proposed detector, due to its data-driven characteristic, provides a generally robust and effective performance for all cases of this inclusive BHT. This appropriate performance of CNN was observed not only for different distributions inside a family of distribution (such as different $S\alpha S$ distributions

generated with different α) but also for different families of distributions like MCA, $S\alpha S$, and CGGD. Additionally, for each of these models, we derived the related detectors based on the model-dependent and highly regarded method of LRT. In simulation results, we indicated that in contrast to the proposed CNN, solutions offered by LRT and other state-of-the-art methods cannot provide uniformly efficient performance in all these models or for signals with large SNR (GSNR) which are not observed by the network in the learning stage. Their performance fluctuates by change of parameters of a distribution, change of the family of distribution. Therefore, for SS problem, our proposed model-free CNN with suitable architecture and its uniformly appropriate performance for all models is an alternative solution to the well-known LRT and other model-based detectors. It also outperforms over the DNN-based detectors compared in this paper. However, the proposed CNN requires higher computational complexity in comparison with these detectors.

APPENDIX A PROOF OF PROPOSITION 1

For proof of Proposition 1, we use the CF of a real with $X \sim \mathcal{N}(\mu, \sigma^2)$ given by [45]

$$\Phi_X(u) = \int e^{iux} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = e^{iu\mu - \frac{\sigma^2 u^2}{2}}. \quad (41)$$

Now, for $r_m[n]|H_1$ with $r_m[n]|H_1 \sim \mathcal{MCA}(A, \Xi, \sigma_w^2, s_m[n])$, the CF by use of (9) is given

$$\begin{aligned} \Phi_{r_m[n]|H_1}(u_{m,n}) &= \iint e^{i\text{Re}[u_{m,n}^H r_m[n]]} \sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p! \pi \sigma_p^2} e^{-\frac{|r_m[n] - s_m[n]|^2}{\sigma_p^2}} \\ &\quad \times dr_m^I[n] dr_m^Q[n]. \end{aligned} \quad (42)$$

By taking integrals inside the summation, we have

$$\begin{aligned} \Phi_{r_m[n]|H_1}(u_{m,n}) &= \sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p! \pi \sigma_p^2} \iint e^{i\text{Re}[u_{m,n}^H r_m[n]]} e^{-\frac{|r_m[n] - s_m[n]|^2}{\sigma_p^2}} \\ &\quad \times dr_m^I[n] dr_m^Q[n], \end{aligned} \quad (43)$$

where we can separate the integrals in (44), shown at the bottom of the page. Now, we use (41) for finding these integrals as written in (45), shown at the bottom of the page.

Thus, we have

$$\Phi_{r_m[n]|H_1}(u_{m,n}) = e^{i\text{Re}[u_{m,n}^H s_m[n]]} \sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p!} e^{-\frac{\sigma_p^2 |u_{m,n}|^2}{4}}. \quad (46)$$

To proceed further, we use the fact that PDF of $r_m[n]|H_1$ can be rewritten by 2-dimensional inverse Fourier of $\Phi_{r_m[n]|H_1}(-u_{m,n})$ as follows

$$\begin{aligned} f(r_m[n]|H_1) &= c_f \iint \Phi_{r_m[n]|H_1}(-u_{m,n}) e^{i(r_m^I[n] u_{m,n}^I + r_m^Q[n] u_{m,n}^Q)} \\ &\quad \times du_{m,n}^I du_{m,n}^Q. \end{aligned} \quad (47)$$

By taking the absolute value of (47), since $f(r_m[n]|H_1)$ is real, we can use the property of integral for absolute values to have the inequality in (48), shown at the top of the next page. If we plug (46) into this inequality and simplify the right side, we reach to (49), shown at the top of the next page. For maximizing the left side of (49), we have to equate both sides of this inequality by selecting the optimum estimation for $s_m[n]$. It is easy to show that equality holds for (49) if the complex part of exponential arguments in the left side becomes zero. Thus, we should have

$$-\text{Re}[u_{m,n}^H s_m[n]] + r_m^I[n] u_{m,n}^I + r_m^Q[n] u_{m,n}^Q = 0, \quad (50)$$

which implies $\text{Re}[u_{m,n}^H s_m[n]] = \text{Re}[u_{m,n}^H r_m[n]]$ and $s_m[n] = r_m[n]$. Hence, we can easily conclude that MLE for problem in (10) is $\hat{\mathbf{S}} = \mathbf{R}$, and Proposition 1 is proved. ■

APPENDIX B PROOF OF PROPOSITION 2

We write $L(\mathbf{R}|H_1)$ with the help of (19) as $L(\mathbf{R}|H_1) = \sum_{m,n} \log [\mathcal{F}^{-1}\{\Phi_{r_m[n]|H_1}(-u_{m,n})\}]$, where by use of (16), $\Phi_{r_m[n]|H_1}(-u_{m,n})$ is

$$\begin{aligned} L(\mathbf{R}|H_1) &= \sum_{m,n} \log [\mathcal{F}^{-1}\{\exp(-i\text{Re}[u_{m,n}^H s_m[n]] - \gamma_s^\alpha |u_{m,n}|^\alpha)\}]. \end{aligned} \quad (51)$$

By use of 2-dimensional inverse Fourier transform, the term inside logarithm function (related PDF), denoted by $f(r_m[n]|H_1)$, can be written as (52), shown at the top of the next page. Now, we use the inequality property of integral for absolute values for (52) to have (53), shown at the top of the next page. Since it is a PDF, the right side of inequality

$$\Phi_{r_m[n]|H_1}(u_{m,n}) = \sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p!} \left(\int \frac{1}{\sqrt{\pi \sigma_p^2}} e^{iu_{m,n}^I r_m^I[n]} e^{-\frac{(r_m^I[n] - s_m^I[n])^2}{\sigma_p^2}} dr_m^I[n] \right) \left(\int \frac{1}{\sqrt{\pi \sigma_p^2}} e^{iu_{m,n}^Q r_m^Q[n]} e^{-\frac{(r_m^Q[n] - s_m^Q[n])^2}{\sigma_p^2}} dr_m^Q[n] \right). \quad (44)$$

$$\begin{aligned} \Phi_{r_m[n]|H_1}(u_{m,n}) &= \sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p!} \left(e^{iu_{m,n}^I s_m^I[n] - \frac{\sigma_p^2 (u_{m,n}^I)^2}{4}} \right) \left(e^{iu_{m,n}^Q s_m^Q[n] - \frac{\sigma_p^2 (u_{m,n}^Q)^2}{4}} \right). \end{aligned} \quad (45)$$

$$\begin{aligned} & \iint \Phi_{r_m[n]|H_1}(-u_{m,n}) e^{i(r_m^I[n]u_{m,n}^I + r_m^Q[n]u_{m,n}^Q)} du_{m,n}^I du_{m,n}^Q \\ & \leq \iint \left| \Phi_{r_m[n]|H_1}(-u_{m,n}) e^{i(r_m^I[n]u_{m,n}^I + r_m^Q[n]u_{m,n}^Q)} \right| du_{m,n}^I du_{m,n}^Q. \end{aligned} \quad (48)$$

$$\begin{aligned} & \iint \sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p!} e^{-\frac{\sigma_p^2 |u_{m,n}|^2}{4}} e^{i(-\text{Re}[u_{m,n}^H s_m[n]] + r_m^I[n]u_{m,n}^I + r_m^Q[n]u_{m,n}^Q)} du_{m,n}^I du_{m,n}^Q \\ & \leq \iint \sum_{p=0}^{\infty} \frac{e^{-A} A^p}{p!} e^{-\frac{\sigma_p^2 |u_{m,n}|^2}{4}} du_{m,n}^I du_{m,n}^Q. \end{aligned} \quad (49)$$

$$f_{r_m[n]}(r_m[n]) = \iint e^{-i\text{Re}[u_{m,n}^H s_m[n]] - \gamma_s^\alpha |u_{m,n}|^\alpha + i(r_m^I[n]u_{m,n}^I + r_m^Q[n]u_{m,n}^Q)} du_{m,n}^I du_{m,n}^Q. \quad (52)$$

$$\begin{aligned} |f_{r_m[n]}(r_m[n])| &= \left| \iint e^{-i\text{Re}[u_{m,n}^H s_m[n]] - \gamma_s^\alpha |u_{m,n}|^\alpha + i(r_m^I[n]u_{m,n}^I + r_m^Q[n]u_{m,n}^Q)} du_{m,n}^I du_{m,n}^Q \right| \\ &\leq \iint \left| e^{-i\text{Re}[u_{m,n}^H s_m[n]] - \gamma_s^\alpha |u_{m,n}|^\alpha + i(r_m^I[n]u_{m,n}^I + r_m^Q[n]u_{m,n}^Q)} \right| du_{m,n}^I du_{m,n}^Q = \iint e^{-\gamma_s^\alpha |u_{m,n}|^\alpha} du_{m,n}^I du_{m,n}^Q. \end{aligned} \quad (53)$$

in (53) is real even without absolute value. Thus, the maximum of $f(r_m[n]|H_1)$ can be reached if both sides of inequality (53) become identical. Therefore, by having

$$-\text{Re}[u_{m,n}^H s_m[n]] + r_m^I[n]u_{m,n}^I + r_m^Q[n]u_{m,n}^Q = 0 \quad (54)$$

in (53), equality holds for (53). We can rewrite (54) as $\text{Re}[u_{m,n}^H s_m[n]] = \text{Re}[u_{m,n}^H r_m[n]]$. It implies that by selecting or estimating $s_m[n]$ equal to $r_m[n]$, we can achieve the maximum amount of PDF of $r_m[n]$ ($f_{r_m[n]}(r_m[n])$) and the maximum likelihood of $L(\mathbf{R}|H_1)$, consequently. Hence, for MLE of $\mathbf{S}$, we can conclude that $\hat{\mathbf{S}} = \mathbf{R}$. ■

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